National Research University Higher School of Economics (HSE)

Curriculum

Field of study 01.03.02 Applied Mathematics and Information Science

Educational Programme "Data Science and Business Analytics"

Specializations: "Data Science in Applied Research", "Data Science in Business", "Data Science in Finance"

Implementing unit: Faculty of Computer Science, HSE - Moscow

3 rd, 2025/2026 academic year

APPROVED

06.06.2025

Vice Rector

ROSHCHIN S.Y.

Signed with EDS

Length of Programme: 4 years

Years of Study: 2023/2024 - 2026/2027

Mode of Study: Full Time

Degree: Bachelor's degree

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Progamme				**60,00**	**2 280**	**852**	**224**	**200**	**264**	**192**	
	Minor				**10,00**	**380**	**112**	**28**	**28**	**40**	**16**	
	Flexible minor				**10,00**	**380**	**112**	**28**	**28**	**40**	**16**	
1	Minor	E			10,00	380	112	28	28A	40	16A	Foreign language
	General Courses				**4,00**	**152**	**76**	**12**	**16**	**24**	**24**	
	Optional General Courses				**4,00**	**152**	**76**	**12**	**16**	**24**	**24**	
1	Russian History	C	School of History		4,00	152	76	12	16A	24	24A	Online Course, Foreign language
	Data Culture				**5,00**	**190**	**60**		**2**	**40**	**18**	
1	External Examinations on Digital Skills. Final Level	C	отдел развития цифровых компетенций				2		2A			Foreign language
2	Deep Learning	C	Department of Big Data and Information Retrieval		4,00	152	56			40	16A	Foreign language
3	Data Science, AI and Generative Models Independent Test. Advanced Level	C	отдел развития цифровых компетенций		1,00	38	2				2A	Online Course, Foreign language
	Internship				**8,00**	**304**	**4**	**2**			**2**	
	Project Internship				**5,00**	**190**	**2**				**2**	
1	Programming Project 2	C		F	5,00	190	2				2A	Foreign language
	Professional Internship				**3,00**	**114**	**2**	**2**				
1	Work Experience Internship	C		F	3,00	114						Foreign language
2	Educational Internship	C		F			2	2A				Foreign language
	Major				**33,00**	**1 254**	**600**	**182**	**154**	**160**	**132**	
	Core Professional Block (Major)				**4,00**	**152**	**56**	**28**	**28**			
	Core Professional Block (Major)				**4,00**	**152**	**56**	**28**	**28**			
1	Optimization Methods	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Online Course, Foreign language
	Elective Professional Block (Major)				**29,00**	**1 102**	**544**	**154**	**126**	**160**	**132**	
	Elective Courses				**4,00**	**152**	**68**	**28**		**40**	**28**	

	Elective Courses (Block 1)				**4,00**	**152**	**68**			**40**	**28**	
1	Information Systems	E	Department of Big Data and Information Retrieval		4,00	152	60			60A		Online Course, Foreign language
2	Operations Research and Game Theory	E	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
3	Foundations of Modern Probability	E	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
4	Advanced Statistical Methods	E	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
	Compulsory Courses				**13,00**	**494**	**288**	**84**	**84**	**60**	**60**	
1	Time Series Analysis	C	Department of Big Data and Information Retrieval		4,00	152	120			60	60A	Foreign language
2	Databases	C	Department of Software Engineering		3,00	114	56	28	28A			Online Course, Foreign language
3	Machine Learning 1	C	Department of Big Data and Information Retrieval		3,00	114	56	28	28A			Foreign language
4	Stochastic processes and applications	C	Department of Big Data and Information Retrieval		3,00	114	56	28	28A			Foreign language
	Specializations				**12,00**	**456**	**188**	**42**	**42**	**60**	**44**	
	Educational track "Data Science in Business"				**12,00**	**456**	**202**	**46**	**54**	**60**	**42**	
1	Data Analysis in Business	C	Department of Big Data and Information Retrieval		4,00	152	72	32	40A			Foreign language
2	Research Seminar "Data Analysis in Business"	C	Department of Big Data and Information Retrieval		4,00	152	62	14	14	20A	14A	Foreign language
3	Investment Management for Business	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
	Educational Track "Data Analysis in Applied Research"				**12,00**	**456**	**188**	**42**	**42**	**60**	**44**	
1	Methods of Mathematic Modeling	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language
2	Research Seminar "Data Analysis in Applied Research"	C	Department of Data Analysis and Artificial Intelligence		4,00	152	64	14	14	20A	16A	Foreign language
3	Applied Statistics for Machine Learning	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
	Educational track "Data Science in Finance"				**12,00**	**456**	**188**	**42**	**42**	**60**	**44**	
1	Research Seminar "Data Science in Financial Economics"	C	Department of Big Data and Information Retrieval		4,00	152	64	14	14	20A	16A	Foreign language
2	Risk management in a bank	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
3	Financial Mathematics	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language

Curriculum agreed:

Academic Supervisor	Voznesenskaia T.V.	04.06.2025
Dean	ARZHANTSEV I.V.	05.06.2025
Head of Centre for Educational Model Design	LEPESHKIN I.A.	06.06.2025

* Subject type:	
Compulsory course	C
Elective course	E

Period of execution:

Fixed	F